INCH-POUND

A-A-52049
January 16, 1992

# COMMERCIAL ITEM DESCRIPTION

## TRIPOD, SURVEYING: FOR RANGE POLE

The General Services Administration has authorized the use of this commercial item description in preference to MILitary specification MIL-T-11674F.

This description covers a lightweight metal folding tripod used in general surveying operations to support a range pole. It may also be used to support a signal light or other surveying equipment.

Salient characteristics.

1. Design. The surveying tripod shall be as described herein, and as shown in figure 1.

2. Tripod assembly:

   a. Leg assemblies: The tripod assembly requires 3 leg assemblies, 2 of which are identical.

      (1) Leg assemblies 1 and 2. The 2 identical legs are described as follows:

         (a) Legs. Each leg is constructed of aluminum alloy. The legs shall be round tubing 36 inches long x 1-inch diameter, with a wall thickness of 0.065-inch. The ends of the tubing shall be chamfered.

         (b) Socket. Each leg shall be constructed with a 2-inch diameter cast aluminum alloy socket which tappers down to 0.375-inch diameter to hold the leg point.

         (c) Point. Each socket shall be fitted with a point of corrosion resistant steel (CRES) hardened to Rockwell C40-45 to provide the tripod contact point with the ground. The point is pinned into the socket.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and other data which may improve this document should be sent by letter to: Commander, U.S. Army Belvoir Research, Development and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606.

AMSC N/A FSC 6675

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(d) Swivel. Each leg assembly includes a 1 1/8-inch diameter aluminum alloy swivel welded into the top of the leg. The swivel provides a pivot contact point to attach the legs to the tripod head.

(2) Leg assembly 3. This leg is drilled to provide attachments for the strap binding assembly, the bracket extension assembly storage, and the level.

(a) Leg. The third tripod leg assembly is constructed of aluminum alloy tubing 36 inches in length (lg) x 1.00-inch outside diameter (OD) and with a wall thickness of 0.065-inch.

(b) Strap binding assembly. The 15.5-inch long strap binding assembly, used to bind the three legs together when the tripod is not in use, consists of webbing, buckle, and a clip.

(c) Swivel. The third leg assembly includes a 1.125-inch diameter aluminum alloy swivel welded into the top of the leg. The swivel provides a pivot contact point to attach the leg to the tripod head.

(d) Socket. The leg shall be constructed with a 2-inch diameter cast aluminum alloy socket which tapers down to 0.375-inch diameter to hold the leg point.

(e) Point. The socket shall be fitted with a point of CRES hardened to Rockwell C40-45 to provide the tripod contact point with the ground. The point is pinned into the socket.

(f) Rivet and washer. A rivet and washer are provided to secure the binding strap to the leg.

(1) Rivet. The rivet is a solid-universal head, aluminum alloy type.

(2) Washer. The washer is a flat-metal, round, general purpose type, made from anodized aluminum alloy.

b. Bracket, extension assembly. A bracket extension assembly is provided to secure the range pole in a vertical position beside the tripod.

(1) Bracket, extension. The extension bracket is fabricated from a section of aluminum alloy. The over all length of the extension bracket is 9.62 inches. A 1.56-inch diameter hole at one end of the bracket holds the range pole. A 6-inch elongated slot allows the range pole to be extended outward from the tripod.



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shall be freely supported on a pivot at approximately

(2) Grommet. A grommet shall be placed inside the 1.56-inch diameter hole in the extension bracket, to protect the range pole.

c. Screw locking assembly. A screw locking assembly shall be provided to secure the extension bracket to the tripod head.

(1) Screw, locking. The extension bracket locking screw shall be cast aluminum alloy or aluminum alloy wrought. The outside top portion of the locking screw shall have a coarse straight knurl. The outside bottom portion shall have 0.625-11 UNC-2A threads. The top portion shall have one hole 0.156-inch diameter on each side to accommodate the bar.

(2) Bar. A bar, consisting of a spring steel wire, shall be inserted into the center of the top portion of the screw in the holes provided. The bar, bent U shape in the center maintains a spring action within the holes and provides an anchor for the link.

(3) Link. A link shall be inserted on the bar in the top center of the screw. The link provides captive chaining for the locking screw assembly when not in use.

d. Bracket, mounting. The mounting bracket shall be aluminum alloy casting with a tapped hole for a CRES helical coil insert. Three holes shall be provided for bolting the leg assemblies to the mounting bracket.

Note: Three screws, nuts, washers, and safety wire are required to attach the leg assemblies to the mounting bracket.

e. Level, circular, rod. A circular rod level consisting of a circular level vial mounted on a right angle support is provided and secured to the leg assembly with the extension bracket. The level vial mounting shall be adjustable.

f. Screw. A hexagonal head cap screw (finished hexagon bolt), shank drilled for cotter pin, is provided to secure the level and the extension bracket to the leg assembly.

g. Strap clamp. A strap clamp is provided to secure the level and the extension bracket to the leg assembly.

3. Identification. The tripod shall be identified with the manufacturer's name or trade mark.

4. Treatment and painting. The tripod shall be cleaned, treated, and painted.



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5. Workmanship. All parts, components, and assemblies of the tripod shall be clean and free from sand, dirt, fins, pits, sprues, scales, flux and other harmful extraneous material.

Certification of commercial item. The contractor shall certify that the product offered to the Government meets the salient characteristics of this description and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices and is the same product as the manufacturer's current product offered by the contractor in the commercial marketplace for at least one year preceding the solicitation. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this commercial item description are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Quality assurance: The tripod shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

1. Design not as specified.
2. Material not as specified.
3. Identification not as specified.
4. Treatment and painting not as specified.
5. Workmanship not as specified.
6. Packaging and marking not as specified.

Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing the contractor's employees to undue hazards from the recovered materials.

Preservation, packaging, packing, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.



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CID based part identification number. The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.

Part or identifying number. The part or identifying number for the tripod, surveying, for range pole shall be as follows:

A 52049

CID Document identifier ——————┘

Notes: The procuring agency should specify the preferred options permitted herein and include the following information in procurement documents:

1. Title, number, and date of this item description.
2. When the commercial item description is used for acquisition, the commercial item certification clause must appear in the solicitation.
3. The tripod must be compatible for use with NSN 6675-00-244-0446 Level, Level Rod.

FED-STD-376 is available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

The following are suggested sources of supply for this item description (Competition is not limited to these sources):

1. Brunson Instrument Company, 8000 East 23rd Street, Kansas City, MO 64129. Telephone: (816) 483-3187.
2. Seiler Instrument & Manufacturing Co., Inc., 170 E Kirkham Ave., St. Louis, MO 63119. Telephone: (314) 968-2282.

The preparing activity for this item description is the US Army Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606

MILITARY INTEREST:

Custodians
Army - ME
Navy - YD

Review activity
DLA - GS

User Activity
Navy - MC

CIVIL AGENCY COORDINATING ACTIVITY:

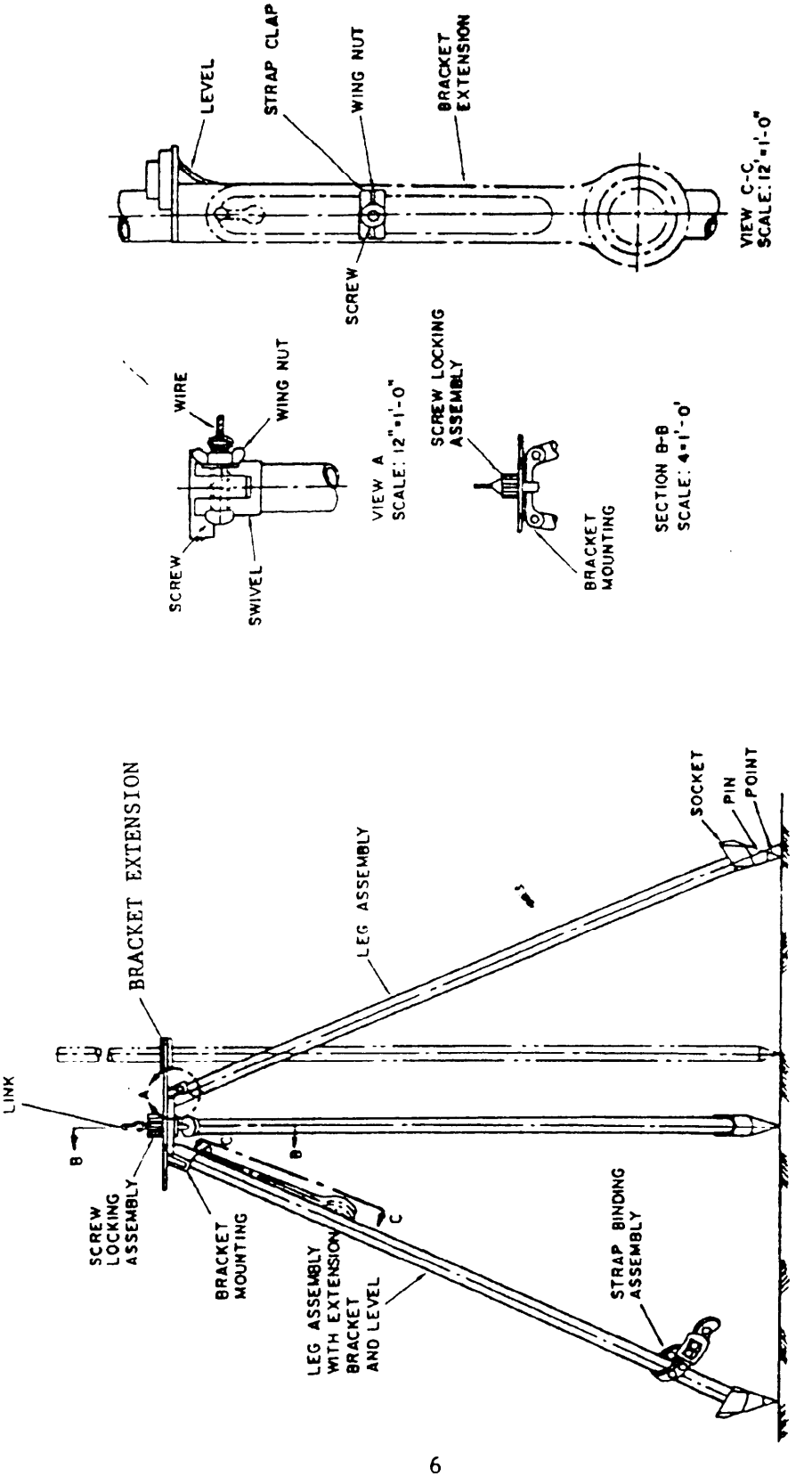
GSA - FCCC

PREPARING ACTIVITY:
Army - ME

Project 6675-0898



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FIGURE 1. Tripod, surveying for range pole.



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